

A Filtering Method for Classification of Motor-Imagery EEG Signals for Brain-Computer Interface

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Abstract—A brain-computer interface (BCI) utilizes brain signals such as electroencephalogram (EEG) and provides a path way for people to interact with external assistive devices. The objective of this work is to classify the tasks so that we can assist the disabled person in doing things on own way with the aid of BCI. The raw EEG signals have a chance of being affected with interference and hence have low signal to noise ratio (SNR) which may lead to erroneous results. These EEG signals are decomposed into intrinsic mode functions (IMFs) using different standard algorithms like empirical mode decomposition (EMD), multi variate empirical mode decomposition (MEMD). Different features like skewness, K-Nearest Neighbour (K-NN) entropy, sample entropy and permutation entropy are extracted from these IMFs which will significantly contribute to the classification of tasks. This work is carried out on the well established BCI motor imagery dataset, BCI competition IVa dataset-1 which will support the analysis. These extracted features are subjected to classifiers like random forest, Naive Bayes and J48 classifiers. The classification accuracies have been recorded and improved results are achieved using MEMD.

Keywords— Brain computer interface (BCI); Motor imagery; EMD; MEMD; Classification; EEG signals

I. INTRODUCTION

Brain-Computer Interface (BCI) is a developing technology whose intention is to pass on an individual's thought processes to the external world straight forwardly from their thinking process and thereby improving cognitive capacities. It is a direct communication channel between the mind and an outside assistive gadget [1]. The most commonly used method in the designing of a BCI system relying on the usage of electroencephalogram (EEG). An EEG signal is used to measure neurophysiological activity [2] in terms of electrical signals generated in the brain. Motor imagery [3] is a mental simulation of the task. Wolpaw et al. translated [1] the neuronal activities into real-time commands developing new

augmentative communication and control technology which is not only concerned with the field of disability but also integrates different domains.

EEG based motor imagery classification is coupled with certain difficulties such as noise addition [4] while recording, due to artifacts caused by muscular activity and poor electrode contact. EEG signals are subject specific, non-stationary, and are greatly prone to noise due to artifacts [5], which makes them difficult to analyze. Gaur et al. [6] proposed a novel preprocessing technique that automatically reconstructs the EEG signal by selecting the intrinsic mode functions (IMFs) based on median measure of frequency. The EEG signal which is reconstructed has shown high signal to noise ratio (SNR) and consists of only that information which is correlated with particular motor imagery task.

All the traditional signal decomposing techniques like short time Fourier transform [7], wavelet transform [8] etc., decompose a signal based on the assumption of linearity, leading to sub-optimal localization, resulting in inadequate performance. This has led to the development of a new data-driven adaptive technique known as, empirical mode decomposition (EMD) [9], which has been established with acceptable performance in majority cases of nonlinear, multi-scale and real-time non-stationary data series. In order to cater the need of real-time multichannel series of data, Mandic et al. [10] developed an extension to EMD algorithm, known as multivariate EMD (MEMD).

A comparative study [5] which analyzes the accuracy of MEMD in dealing with non-stationarity, closely spaced frequency bands and low SNR is made for BCI competition IV data set-1. This study recorded maximum accuracy of 91.5% for subject-G. In [11], Suman Dutta used multivariate auto regressive models for six channel EEG data in MEMD

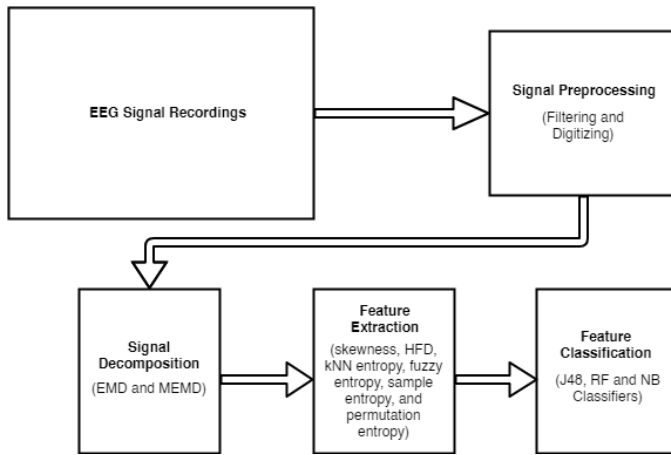


Fig. 1. Classification system for BCI Application

domain. Eigenvalues calculated from the co-variance matrices have shown a high distinguishing accuracy of 94.3%. Gaur [12] analysed the features that are extracted utilizing single and multichannel EMD based filtering for the classification of the multiple direction wrist movements based on Magnetoencephalography (MEG) signals for improving BCI. This analysis proved that the multichannel filtering can be utilized in online BCI systems for effective results. Fig.1 shows the classification system for BCI application.

The objective of this work is to develop a filtering method based on a new methodology for BCI motor-imagery classification using EEG signals which can be easily deployed in an effective computing framework thereby providing disabled human beings with a new communicate channel.

The remaining of this paper is organized as below. Section II contains a brief overview of EMD. Section III and IV describe an outline of MEMD and detailed implementation of our method, respectively. In section V the results for different decomposition techniques are discussed and finally section VI contains conclusion.

II. EMPIRICAL MODE DECOMPOSITION

The EMD [9] has been proposed as a data adaptive method. In the EMD approach, data which is represented as $Y(x)$ is decomposed into number of IMFs [10]. IMFs are simple oscillatory mono-component functions with a varying frequency and amplitude. They are of the identical length as the source signal and conserve the frequency alterations with time. The signal can be reconstructed by simply summing up the IMFs as mentioned in (1) as follows:

$$Y(t) = \sum_{i=0}^n IMF_i + k_n \quad (1)$$

Where k_n is the residue of the source signal $Y(t)$ and n is the number of extracted IMFs.

IMFs have the following properties:

(a) In the entire data set, the total number of zero crossings and the number of extremas must be either differ by at most

one or be equal.

(b) At any instant, the mean of the envelope formed by local minima and local maxima must be equal to zero.

A. The EMD algorithm

In EMD implementation, sifting process which uses local extrema is applied. For the signal $Y(t)$, EMD algorithm is summarized as follows:

Step 1) Detect all local extrema (a combination of maximas and minimas) of the signal $Y(t)$.

Step 2) Connect all these local maximas (minimas) with corresponding splines based on the application and generate the upper (lower) envelope $l_u(t)$ ($l_i(t)$).

Step 3) Obtain the local mean $me(x)$ of the lower and upper envelopes using following expression:

$$me(t) = [l_u(t) + l_i(t)]/2$$

Step 4) Consider $de(t) = Y(t) - me(t)$ as the data, and reiterate steps 1 and 2 multiple times until the normalized squared difference between two consecutive sifting operations defined as in equation (2) to be minimal.

$$SD_i = \frac{\sum_{i=0}^T |de_{i-1}(t) - de_i(t)|^2}{\sum_{i=0}^T de_{i-1}(t)^2} \quad (2)$$

If this squared difference SD_i is lesser than the threshold value, the process will be terminated and can be treated as sufficient IMF and the equivalent residue $r(t)$ is computed by

$$r(t) = Y(t) - de_k(t) \quad (3)$$

III. MULTIVARIATE EMPIRICAL MODE DECOMPOSITION

Though it has been proved that EMD is very useful and data-adaptive, it still leaves behind some aggravating difficulties unresolved. Few major shortcomings of EMD are mode mixing, non-uniqueness (different number of IMFs for different channels) and mode misalignment (absence of synchronization and alignment of frequency information among different modes). In addition to this, employing EMD to each and every channel of the multivariate signal is tedious and generates sub-optimal results due to the drawbacks stated. A multivariate extension of EMD is proposed for alignment of scale among different channels.

MEMD [13] is a comprehensive extension of EMD for signal analysis containing multiple channels, by directly functioning in multidimensional space of actual signal. In MEMD, the critical task of finding the local mean of a multivariate signal is done through the multiple real-valued projections of the actual signal along the direction vectors. The direction vectors of multidimensional space are chosen by using uniform angular coordinates and low-discrepancy point sets that arise from quasi-Monte Carlo methods.

The MEMD algorithm can be explained with the following steps:

Step 1) Choose suitable direction vectors $d^{(n)}$ which are samples (point set) on an $M-1$ sphere.

TABLE I
BCI COMPETITION IV-DATASET 1: SUBJECTS AND NUMBER OF TRIALS
FOR RESPECTIVE TASKS.

Subject	Left task	Right Task	Foot Task
A	100	0	100
B	100	100	0
C	100	100	0
D	100	100	0
E	100	100	0
F	100	0	100
G	100	100	0

Step 2) Obtain the projections of the signal $y(t)_{T=1}^X$ along the chosen multiple direction vectors for all L (where L is the total number of direction vectors), resulting in $pr^{\theta_l}(t)_{l=1}^X$ as projections.

Step 3) Note the time points $x_K^\theta(t)$ that are correlated with the maxima of each of the projected signals

Step 4) Interpolate $x_L^\theta(t)$, $y(t)_{t=1}^X$ using a corresponding spline and generate multivariate envelope set $E^{\theta_L}(t)_{L=1}^L$

Step 5) For a total number of L direction vectors, the mean $me(t)$ of the envelope set is calculated as

$$me(t) = \frac{1}{L} \sum_{k=1}^L E^{\theta_k}(t)$$

Step 6) Consider $de(t) = y(t) - me(t)$ as the new data and if $d(t)$ fulfils the stopping criterion for an IMF, re-iterate above procedure to $de(t) - y(t)$ else re-iterate for $de(t)$.

IV. OUR PROPOSED APPROACH

A. BCI IV Dataset 1

The data [14] was taken from seven healthy subjects where three motor imagery actions are performed without feedback. The three motor imagery tasks are left, right and foot. For a specific subject only two motor imagery actions were chosen and recordings were continuously taken by switching subjects intentions randomly. Each subject performs a total of 200 trials with a 100 trials of each task as shown in Table 1.

In one trial subject has imagined about one task and the length of one trial is about 4s. The data was taken from 59 EEG channels using a Ag/AgCl electrode cap and BrainAmp MR plus amplifiers. Out of these 59 channels, 11 channels were selected from the central part of the brain where the brain activity is maximum. They are FC3, FC4, Cz, C3, C4, C5, C6, T7, T8, CCP3 and CCP4 in accordance with the 10-20 system. In this dataset the preprocessing of signals is done in two steps. They are segmentation based on task and filtration. The data of recordings was sampled at 1000 Hz initially and then downsampled to 100 Hz.

B. Features

The accuracy of the classification is decided by the features choosen to distinguish the signals [15]. Following features are used in this work:

1) *Skewness*: Skewness is a measure of the proportion of asymmetry present in statistical distribution, where the curve seems misshaped either to the left or to the right. It can be used to describe to the degree to which a distribution contrasts from a normal distribution. It can be expressed mathematically as follows:

$$skewness = \sqrt{\frac{\sum_{j=1}^n (\bar{t} - t_j)^3}{(n-1)s^3}}$$

$\bar{t}$ = Mean

s = Standard Deviation

n = Sample Size

2) *K-NN entropy*: K-NN entropy depends on statistics of k-nearest neighbor (kNN) distances [16]. Let the probability density $\phi_k(\epsilon)$ of ϵ , the distance from x_i to its kNN. The density function $\phi_k(\epsilon)d\epsilon$ indicates the probability of strictly one point which is in $[\epsilon, \epsilon + d\epsilon]$, strictly $k-1$ points at a distance less than the kNN, and the remaining points farther than the kNN. It follows as:

$$\phi_k(\epsilon)d\epsilon =$$

$$\binom{N-1}{1} \frac{d\phi_i(\epsilon)}{d\epsilon} d\epsilon \binom{N-2}{k-1} (\phi_i(\epsilon))^{k-1} (1 - \pi_i(\epsilon))^{N-k-1}$$

The expected value of $\log(\Pi)$ can be obtained as

$$E(\log \phi_i) = \int_0^\infty \log \phi_i(\epsilon) \phi_k(\epsilon) d\epsilon = \psi(k) - \psi(N)$$

$$\phi_i \approx \eta_i \phi_x(x_i)$$

$$\hat{H}(X) = \psi(N) - \psi(k) + \frac{1}{N} \sum \log \eta_i$$

C. Sample entropy

It is utilised to compute the regularity in data without knowledge about a system. Initially, the vectors are constructed in a pseudo-phase: from a time series of N points $se(1), se(2), \dots, se(N)$, k is the embedding dimension [17].

$$C_i^k(r) = \frac{(\text{number of } t(j) \text{ such that } d[t(i), t(j)] \leq r)}{N - k + 1}$$

And r is a level of filtering, while d indicates distances of points. Generally, $r = 20\%$ of the standard deviation of the magnitude values and $k = 2$.

$$\phi^k(r) = \frac{1}{N-k} \sum_{i=1}^{N-k} C_i^k(r)$$

Similarly, $k+1(r)$ can be defined as an embedding dimension of $k+1$. The Sample entropy is computed as:

$$Samp_{en} = \ln \frac{\phi^k(r)}{\phi^{k+1}(r)}$$

D. Fuzzy entropy

It is almost similar to the related measure of sample entropy but the concept of fuzzy sets is imported on the basis of exponential functions [18].

In calculating fuzzy entropy [17], initially the vectors constructed in the pseudophase are transformed as follows:

$$[\bar{t}(i)] = [se(i) - \bar{se}(i), \dots, se(i+p-1) - \bar{se}(i)] \quad (4)$$

$$[\bar{t}(j)] = [se(j) - \bar{se}(j), \dots, se(j+p-1) - \bar{se}(j)] \quad (5)$$

where $\bar{x}(i)$ is the mean value of $t(i)$:

$$\bar{se}(i) = \sum_{k=0}^{p-1} \frac{se(i+k)}{p}$$

In the next step, the fuzzy membership matrix is defined as:

$$D_{i,j}^p = \mu(d(\bar{t}_i^p, \bar{t}_j^p))$$

with the fuzzy membership function:

$$\mu(se) = e^{-\left(\frac{t}{r}\right)^n}$$

Finally, fuzzy entropy is defined as:

$$Fuzzy_{en} = \ln \frac{\phi^p}{\phi^{p+1}}$$

where

$$\phi^p = \frac{1}{N-p} \sum_{i=0}^{N-p} \sum_{j=1, j \neq i}^{N-p} \frac{D_{i,j}^p}{N-p-1}$$

E. Permutation entropy

The permutation entropy provides a method to quantify the complexity of time series [19]. Basically, each time series is associated with a probability distribution ϕ , whose elements ϕ_j are the frequencies connected with j feasible permutation patterns, where $i = 1, \dots, n!$. Permutation entropy (P_{en}) is defined as

$$P_{en} = - \sum_{j=1}^{n!} \phi_j \ln \pi_j$$

Normalized permutation entropy is defined as:

$$P_{enorm} = - \frac{1}{\log_2 n!} \sum_{j=1}^{n!} \phi_j \log_2 \phi_j$$

F. Classifiers

1) *J48 Classifier*: J48 is a decision tree algorithm which estimates the target variable class with the help of an ID3 tree classification approach [18]. Besides the simple and comprehensible distribution of data, J48 consists the additional features such as decision trees pruning, accounting for missing values, the abstraction of rules, etc. The aim is a continuous generalization until it reaches equilibrium in its accuracy as well as the flexibility of the tree.

2) *Naive Bayes Classifier (NB)*: Naive Bayes is one of the easy technique for building classifiers based on a basic principle i.e it assumes that a specific feature is independent of other features when class attributes are given [20]. In spite of its amateur design and oversimplified hypothesis, this classifier has given satisfactory performance in case of complex real-world situations. The benefit of Naive Bayes is that this classifier requires a relatively less training set to determine the criterion which is essential for classification.

3) *RF Classifier*: RF classifier comprises a collection of structured decision tree classifiers $t(a, \theta_n)$, $n = 1, \dots, k$ where θ_n are random vectors which are distributed independently. Every individual tree determines a poll for the common output class for given input x [21].

For a assemblage of trees $t_1(a), t_2(a), \dots, t_n(a)$, and with the help of a training data set which is taken randomly from the random vector distribution R, a , the limiting function is detailed as

$$\text{Lmt}(A, R) = av_n \ln(t_n(A) = R) - \max_{j \neq R} av_n I(t_n(A) = j)$$

where $\ln()$ is the index function.

The limit indicates the degree by which the mean number of votes at A, R for the right class surpasses the average decision in favour of some other task.

V. RESULTS AND DISCUSSION

The dataset used in this study consists on average of 190518 data points from 59 channels comprising all the 7 subjects. In this work, we have extracted four seconds duration EEG epochs from each of the 11 EEG channels for seven different subjects. This results in 30800 number of left EEG epochs, 22000 number of right EEG epochs, and 8800 number of foot EEG epochs for motor imagery classes. For all the IMFs, 5 features had been extracted which are K-NN entropy, fuzzy entropy, sample entropy, permutation entropy, and skewness. A correlation-based feature selection filter has been applied on the extracted features. Percentage split of 50% is used as criteria for the splitting of J48, RF and NB. . The extracted IMFs using MEMD and EMD of the three motor imagery classes of EEG signals are shown in Figures 2, 3, 4, 5, 6, and 7 respectively.

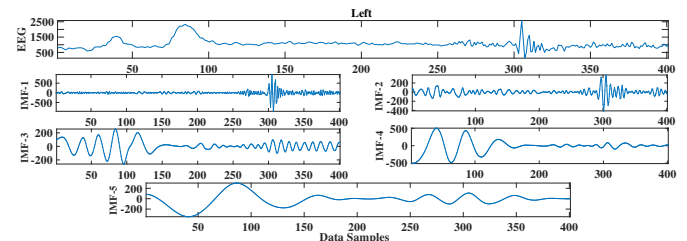


Fig. 2. Plot of EEG signal of FC3 channel and its extracted IMFs using EMD method for left motor imagery task of subject A

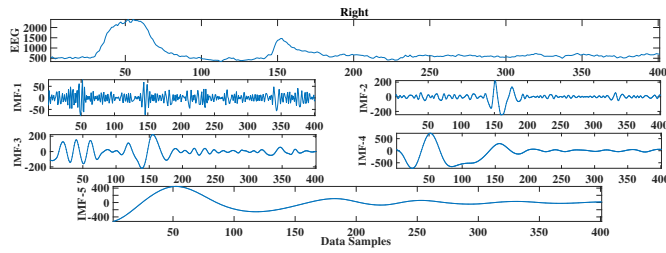


Fig. 3. Plot of EEG signal of FC3 channel and its extracted IMFs using EMD method for right motor imagery task of subject A

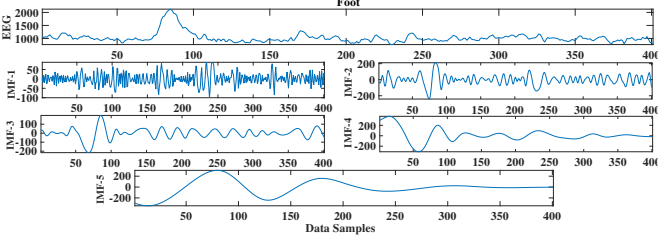


Fig. 4. Plot of EEG signal of FC3 channel and its extracted IMFs using EMD method for foot motor imagery task of subject A

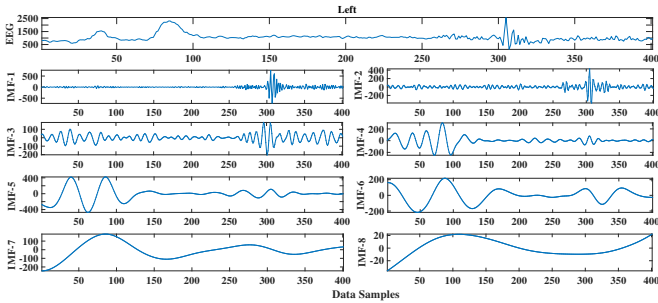


Fig. 5. Plot of EEG signal of FC3 channel and its extracted IMFs using MEMD method for left motor imagery task of subject A.

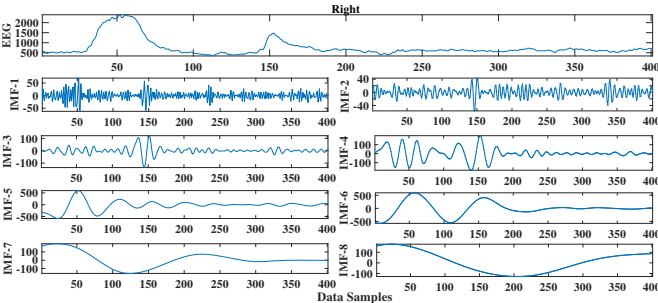


Fig. 6. Plot of EEG signal of FC3 channel and its extracted IMFs using MEMD method for right motor imagery task of subject A.

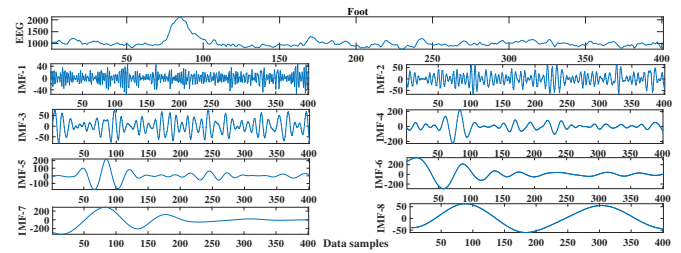


Fig. 7. Plot of EEG signal of FC3 channel and its extracted IMFs using MEMD method for foot motor imagery task of subject A.

The IMFs obtained in the case of EMD are downsized to a 5 minimum number of IMFs which are common to different epochs. Whereas in MEMD, the problem of non-uniqueness is eliminated and an identical number of 8 IMFs are obtained. It is evident from the IMFs of both EMD and MEMD that the rapid changes in the signal are captured first followed by the slow changes and each IMF is of the same size as that of the original EEG signal.

After decomposition, feature extraction (K-NN entropy, fuzzy entropy, sample entropy, permutation entropy, HFD, and skewness) is performed for all the 7 subjects. As mentioned earlier, 6 features are computed from each IMF resulting in 30 features per channel in case of EMD and 48 features per channel in case of MEMD. Therefore the dimension of feature vector for EMD and MEMD are $R^{30Cn \times 1}$ and $R^{48Cn \times 1}$ respectively, where Cn indicates the number of channels.

The Figures 9-11 show the box plots of IMF-1 for the channel FC3 corresponding to sample entropy, K-NN entropy, and fuzzy entropy features. It is evident from the figures that there is significant discrimination among three different classes for most of the features.

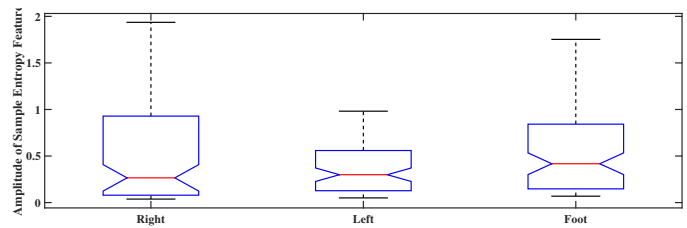


Fig. 8. Box plots obtained for right, left, and foot motor imagery tasks using sample entropy feature for channel FC3 for IMF 1.

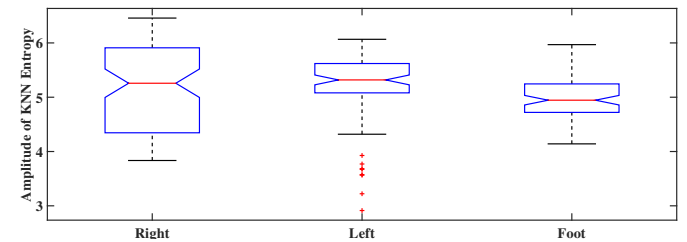


Fig. 9. Box plots obtained for right, left, and foot motor imagery tasks using KNN entropy feature for the channel FC3 for IMF 1.

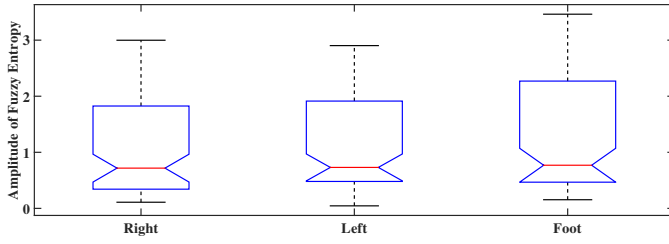


Fig. 10. Box plots obtained for right, left, and foot motor imagery tasks using fuzzy entropy feature for channel FC3 for IMF 1.

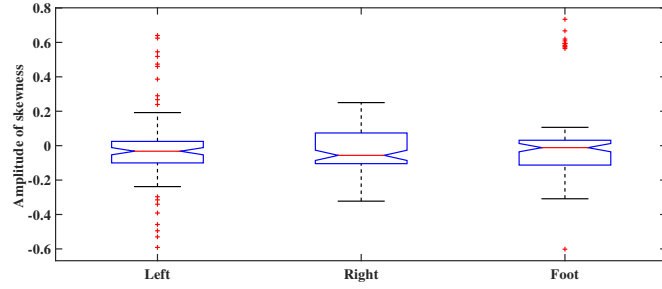


Fig. 11. Box plots obtained for right, left, and foot motor imagery tasks using skewness feature for channel FC3 for IMF 1.

In order to reduce the dimensions of feature vector, a filter called correlation based feature selection (CFS) [22] is applied on the feature matrix. The processed features are then fed to classifiers (RF, NB, and J48) using percentage split (50% of the feature vector as training and test datasets). A comparison of obtained classification accuracies for the extracted features using the 3 classifiers is presented in Table II.

The detailed performance of MEMD for each individual subject is evaluated using 10 fold cross-validation [23] and percentage split are presented in Tables III, IV,

There is a considerable increase in the classification performance using MEMD technique as compared to EMD technique. The maximum reported accuracies for subjects B, C, D, and E are obtained using MEMD technique along with RF classifier and the accuracies are 93%, 98%, 99.5%, and 99% respectively. Similarly, in the case of remaining subjects, accuracies are improved using MEMD. Among the three

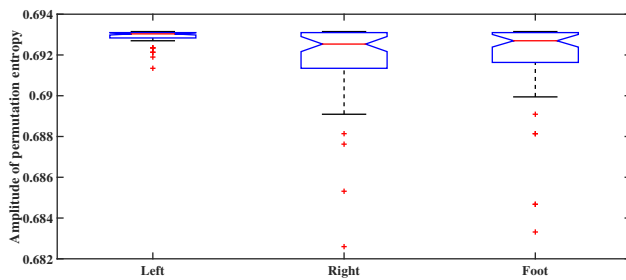


Fig. 12. Box plots obtained for right, left, and foot motor imagery tasks using permutation entropy feature for channel FC3 for IMF 1.

TABLE II
CLASSIFICATION ACCURACY FOR J48, RF AND NB CLASSIFIERS FOR 11 CHANNEL DATA WHEN MEMD AND EMD ALGORITHMS ARE USED.

Subjects	J48 Decision Tree		RF		NB	
	EMD	MEMD	EMD	MEMD	EMD	MEMD
Subject A	54%	57 %	55%	58%	54%	56%
Subject B	46%	84.5 %	44%	93%	51%	68.5%
Subject C	49 %	89%	52%	98%	51%	94.5%
Subject D	47%	94.5%	41%	99.5%	49%	95.5%
Subject E	54%	92 %	55%	99%	46%	81.5%
Subject F	50 %	69%	50%	72 %	51%	66.5%
Subject G	49 %	64%	55 %	74 %	52%	65%

TABLE III
CLASSIFICATION ACCURACY FOR J48, RF AND NB CLASSIFIERS FOR 7 SUBJECTS WHEN MEMD ALGORITHM IS USED ALONG WITH CFS FILTER.

Classifiers	Cross Validation			Percentage Split		
	J48	RF	NB	J48	RF	NB
Subject A	52%	52.5%	56%	66.5%	52.5%	59.5%
Subject B	84.5%	93 %	68.5%	86.5%	94%	92.5%
Subject C	89 %	98%	94.5%	92%	98%	98.5%
Subject D	94.5%	99.5%	95.5%	94.5%	99.5%	99.5%
Subject E	92%	99 %	81.5%	92.5%	99%	99.5%
Subject F	69 %	72%	66.5%	76.5 %	79.5%	73.5%
Subject G	64 %	74%	65 %	71.5 %	77%	68.5%

classifiers the reported average accuracies for all the seven subjects using J48, RF, and Naive Bayes are 78.7%, 84.8%, and 75.3% respectively. From this we can rank the efficiency of classification of the three classifiers used in our study with RF having the best accuracy followed by J48 and Naive Bayes. The average accuracies across all the subjects using percentage split (88.78%) is comparatively higher than the same reported for 10 fold cross validation (83.65%). It is evident from the tables that the application of CFS filter for most of the cases has provided improved classification accuracy. This resulted into average classification accuracies of 87.3% and 85.3% for with and without the application of CFS filter. Thus it can be stated that the application of MEMD and CFS filter has enhanced the overall performance of the proposed technique for most of the considered subjects.

VI. CONCLUSIONS

A good performance accuracy has always been a challenge in BCI motor imagery domain. In this study we have developed a combination of adaptive signal processing inspired from the

TABLE IV
CLASSIFICATION ACCURACY FOR J48, RF AND NB CLASSIFIERS FOR 7 SUBJECTS WHEN MEMD ALGORITHM IS USED WITHOUT CFS FILTER.

Classifiers	Cross Validation			Percentage Split		
	J48	RF	NB	J48	RF	NB
Subject A	54%	55%	62%	58%	59%	58%
Subject B	86%	91 %	59%	87%	93%	93%
Subject C	96 %	98%	89%	92%	96%	99%
Subject D	90%	99.5%	93%	91%	99.5%	99.5%
Subject E	85%	99 %	78%	94%	99.5%	99.5%
Subject F	62 %	71%	61%	71.5 %	77%	72%
Subject G	67 %	70%	59 %	65 %	74%	59%

empirical methods (EMD, and MEMD) along with a proper feature set (K-NN entropy, fuzzy entropy, sample entropy, permutation entropy, and skewness) which yielded an optimal performance for 4 out of 7 subjects. This method outperforms the existing work carried out on this database.

In the future extension of work, it may intrigue to assess the proposed technique for classification issue in real-time BCI and help the people. Furthermore, to enhance the proposed technique to characterize the EEG into multiple (more than two) classes. In addition to this, a conventional algorithm which can lower the number of EEG channels without trading off the classification accuracy can also be developed in the future.

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